CRT questions

1. Decision Structures

a) Display Great job! when grade is 90 or higher:

```
if (grade >= 90) {
    System.out.println("Great job!");
}
```

b) Display Error when number is less than 20 or greater than 50:

```
if (number < 20 || number > 50) {
    System.out.println("Error");
}
```

c) Add 2 to the value of y when y is less than 100:

```
if (y < 100) {
    y += 2;
}
```

2. Number Comparison if-else if

Assume num1 and num2 contain integer values. Write an if-else if statement that displays the appropriate message:

```
if (num1 > num2) {
    System.out.println("First number is larger.");
} else if (num2 > num1) {
    System.out.println("Second number is larger.");
```

```
} else {
    System.out.println("Numbers are equal.");
}
```

3. Modulus and Switch Statements

a) Which is the appropriate word, odd or even, for the blanks below?

- First blank: `even`
- Second blank: `odd`

b) Rewrite the if-else as a switch statement:

```
switch (num % 2) {
    case 0:
        System.out.println("even number");
        break;
    default:
        System.out.println("odd number");
        break;
}
```

4. Random Numbers (Math.random())

a) Generate a random integer between 1 and 50:

```
int randNum = (int)(Math.random() * 50) + 1;
```

b) Generate a random integer between 20 and 100:

```
int randNum = (int)(Math.random() * 81) + 20;
```

c) Generate a random double between 10 and 20, inclusive:

```
double randNum = Math.random() * 10 + 10;
```

5. Identifying Logic Errors

Explanation: The original code uses separate if statements instead of an if-else if chain, causing overlapping or incorrect evaluations.
Corrected vers:

```
if (age < 18) {

   System.out.println("child");

} else if (age <= 65) {

   System.out.println("adult");

} else {

   System.out.println("senior");

}
```

6. Boolean Expression Evaluation

Given: `size = 100`, `weight = 50`, `value = 75`:

a) `size > 50 && weight == 50` — **true**
b) `value < 100 && !(weight == 50)`--- **false**
c) `size >= 100 || value >= 100` — **true**
d) `weight < 50 || size > 50` — **true**
e) `!(value < 75)` — **true**
f) `!(size > 100 && weight > 50 && value > 75)` — **true**
g) `(value < 125 || weight < 76) && size == 100` — **true**

8. True/False Questions

a) The condition of an if statement must be a Boolean expression. — **True**

b) A nested if statement and an if-else statement are the same. — **False** (A nested if is placed inside another block, whereas an if-else provides an alternative branch)
c) The expression in a switch statement must evaluate to a double. — **False** (Switch expressions evaluate to integers, strings, characters, or enums, never doubles)
d) Numbers generated by a computer program are actually pseudorandom. — **True**
e) The (double) cast is needed to generate a random integer.--- **False** (Casting to int is used for integers)
f) A compound Boolean expression can contain more than two Boolean expressions. — **True**
g) In a logical And expression, both operands must be true for the expression to evaluate to true. —**True**
h) In logical expressions, && is evaluated before `!`. — **False** (The logical NOT operator `!` has higher precedence)
i) The `pow()` method in the Math class is used for exponentiation. — **True**
j) The statement `x = abs(-3);` will return the value 3. — **False** (Must be written as `Math.abs(-3))`